

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Supplementary Summer – 2023 Course: B. Tech. Branch : Mechanical Semester : VI Subject Code & Name: BTMEC603 Applied Thermodynamics-II Max Marks: 60 Date:18/07/2023 Duration: 3 Hr.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly. 5. Use of steam table is permitted			
		(Level/CO)	Marks
Q. 1	Solve Any Two of the following.		12
A)	How calorific value of a fuel is calculated with the help of Bomb calorimeter?	CO 1	6
B)	5 kg coal burnt completely in a furnace. Calculate amount of carbon dioxide generated. How much carbon monoxide will be produced if incomplete combustion takes place?	CO 1	3+3
C)	If a fuel contains 86% carbon, 11.75% hydrogen and 2.25% oxygen, calculate theoretical air supplied per kg of fuel and the weight of products of combustion.	CO 1	3+3
Q.2	Solve Any Two of the following.		12
A)	Enlist six requirements of a good boiler?	CO 1	6
B)	Explain Boiler efficiency and equivalent evaporation.	CO 1	3+3
C)	Feed water having enthalpy of 142 kJ/kg heated in a boiler till it becomes 95% dry at 11.5 bar ($h_f = 790$ kJ/kg and $h_{fg} = 1991$ kJ/kg). Calculate boiler efficiency if 8 kg mass of water evaporated per kg of coal having CV equal to 29,000 kJ/kg .	CO 1	6
Q. 3	Solve Any Two of the following.		12
A)	Draw PV and TS diagram of ideal Rankine cycle and explain each process.	CO 3	2+2+2
B)	The adiabatic enthalpy drop across the prime mover of the Rankine cycle is 840 kJ/kg. The enthalpy of steam supplied is 2940 kJ/kg. and at back pressure of 0.1 bar. Calculate the specific steam consumption and thermal efficiency of the cycle.	CO 2	3+3
C)	Starting from equation of first law, establish relation between exit speed and enthalpy drop in an isentropic nozzle.	CO 2	6
Q.4	Solve Any Two of the following.		12
A)	Why condensers are used in a steam power plant? How they are classified?	CO 3	3+3
B)	Steam enters a 2 MW adiabatic turbine steadily at 30 bar and 400°C and leaves at 0.5 bar and 100°C. Determine the isentropic efficiency of the turbine and the mass flow rate of the steam flowing through the turbine.	CO 3	3+3

C)	The outlet and inlet temperatures of cooling water to a condenser are 37 ⁰ C nd 30 ⁰ C respectively. Calculate condenser efficiency if barometer reads 706 mm of Hg with atmosphere pressure reading 760 mm.	CO 4	6
Q. 5	Solve Any Two of the following.		12
A)	How air compressors are classified?	CO5	6
B)	What are the methods to increase isothermal efficiency of a high speed compressor?	CO5	3+3
C)	A single stage compressor take 1 m ³ air per minute at 1 bar and 15°C and delivers at 15 bar. Assuming $PV^{1.35} = \text{constant}$, calculate indicated power neglecting clearance.	CO5	6
	*** End ***		

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